

ECON 311

Lecture 3: Measuring the Aggregate Economy, part II

Measuring Changes Across Time

Measuring Differences Across Countries

Introduction to Long-Run Growth

GDP across time

We call the sum of the value of production at a particular time (t), nominal GDP:

$$NGDP_t = \sum_{i=1}^N p_{i,t} \cdot q_{i,t} \quad , \quad i = 1, \dots, N, (e.g. \ t = 2018)$$

- $\Delta NGDP$ across time happens because:
 1. Δp_i
 2. Δq_i
- To measure real changes in the economy, we need to separate 2 from 1

Quantity Indexes

Suppose we want to measure real changes between 2018 and 2019, there are two widely used quantity indexes

1. Laspeyres:

$$RGDP_{2018}^{2018\$} = \sum_{i=1}^N p_{i,2018} \cdot q_{i,2018} \quad RGDP_{2019}^{2018\$} = \sum_{i=1}^N p_{i,2018} \cdot q_{i,2019}$$

$$\% \Delta RGDP_L = \frac{RGDP_{2019}^{2018\$} - RGDP_{2018}^{2018\$}}{RGDP_{2018}^{2018\$}} * 100$$

2. Paasche:

$$RGDP_{2018}^{2019\$} = \sum_{i=1}^N p_{i,2019} \cdot q_{i,2018} \quad RGDP_{2019}^{2019\$} = \sum_{i=1}^N p_{i,2019} \cdot q_{i,2019}$$

$$\% \Delta RGDP_P = \frac{RGDP_{2019}^{2019\$} - RGDP_{2018}^{2019\$}}{RGDP_{2018}^{2019\$}} * 100$$

Quantity Indexes

- The only time $\% \Delta RGDP_L = \% \Delta RGDP_P$ is when either all prices or all quantities change proportionally, or both.
- To partially correct for this, a weighted average between the two years, called the Fisher index is implemented:

$$\% \Delta RGDP_F = \frac{\% \Delta RGDP_L + \% \Delta RGDP_P}{2}$$

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	NGDP
2018	1	500	900	5	\$5000
2019	2	600	1000	5	\$6200
2020	3	700	1000	6	\$8100

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$\% \Delta RGDP_L$
2018	1	500	900	5	
2019	2	600	1000	5	2%
2020	3	700	1000	6	19.35%

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$RGDP_L$ Chained Prices benchmarked to 2020
2018	1	500	900	5	\$6650.9
2019	2	600	1000	5	\$6783.9
2020	3	700	1000	6	\$8100

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$\% \Delta RGDP_P$
2018	1	500	900	5	
2019	2	600	1000	5	3.3%
2020	3	700	1000	6	19.12%

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$RGDP_P$ Chained Prices benchmarked to 2020
2018	1	500	900	5	\$6583.7
2019	2	600	1000	5	\$6801
2020	3	700	1000	6	\$8100

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	$\% \Delta RGDP_F$
2018	1	500	900	5	
2019	2	600	1000	5	2.65%
2020	3	700	1000	6	19.24%

Example– Calculating Chained Prices RGDP

Year	P_{apple}	Q_{apple}	$P_{computer}$	$Q_{computer}$	RGDP Chained Weighted Prices benchmarked to 2020
2018	1	500	900	5	\$6616.7
2019	2	600	1000	5	\$6795.3
2020	3	700	1000	6	\$8100

Price Indexes and Inflation

- To calculate price changes across time, we use price indexes:

$$GDP_t^{deflator} = \frac{NGDP_t}{RGDP_t}$$

$$Inflation_t = \frac{GDP_t^{deflator} - GDP_{t-1}^{deflator}}{GDP_{t-1}^{deflator}}$$

Example– Price Indexes and Inflation

Year	NGDP	RGDP Chained Weighted Prices benchmarked to 2020	$GDP^{deflator}$	Inflation Rate
2018	\$5000	\$6616.7	0.756	
2019	\$6200	\$6795.3	0.912	20.63%
2020	\$8100	\$8100	1	9.75%

GDP Across Time– Conclusion

- We use Quantity and Price Indexes to separate the effect of changes in real quantities from changes in prices.
- An alternative and simpler way of calculating quantity and price indexes is to use constant prices for a given year
 - ▶ Introduce problems when used over long periods in cases where
 - ▶ relative prices of goods changes because of changes in their relative quality (e.g. computers)
 - ▶ Chain Weighting solves this issue by both...
 - ▶ considering short time periods (chain)
 - ▶ combining information from both current and future prices (weight)

Measuring Changes Across Time

Measuring Differences Across Countries

Introduction to Long-Run Growth

Differences in Economic Activity Across Countries

- We could compare NGDP using a common currency (e.g. transform Canadian NGDP in US Dollars):

$$NGDP_{CAN}^{USD} = \frac{1}{e_{CAD/USD}} NGDP_{CAN}^{CAD}$$

- ▶ $e_{CAD/USD}$ does not perfectly reflect differences in prices across countries (e.g. differences in production costs)...
 - ▶ so comparing $NGDP_{CAN}^{USD}$ with $NGDP_{US}^{USD}$ has the same problem as comparing $NGDP$ across time
- To control for price differences, we use (PPP)

$$PPP_GDP_{CAN}^{USD} = \frac{P_{US}}{P_{CAN}^{USD}} NGDP_{CAN}^{USD}$$

RGDP Across Countries

	Canada	United States	Mexico
GDP Growth	1.7%	1.7%	1.4%
GDP per head	\$57,190(PPP:\$67,090)	\$86,190(PPP:\$86,190)	\$14,290(PPP:\$27,060)
Inflation	2.2%	3.0%	4.0%
Budget Balance (% GDP)	-1.0	-6.9	-4.0
Population	40.1M	347.3M	131.9M

Measuring Changes Across Time

Measuring Differences Across Countries

Introduction to Long-Run Growth

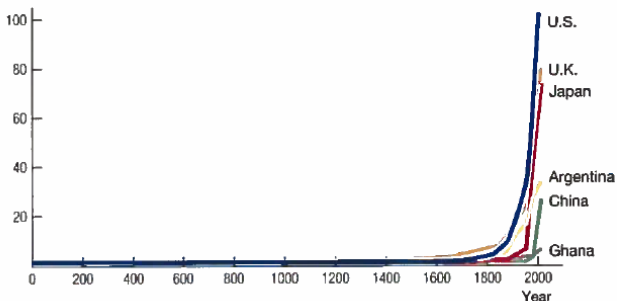
Introduction to Growth

- Standards of living in industrialized countries have greatly increased over the past few centuries.
 - ▶ E.g. Average incomes today in North America and Western Europe are between 50 and 300 times larger than two centuries ago.
- This increase in standards of living has been a very recent phenomena in human history.
 - ▶ For most of their history (before the Industrial Revolution), humans lived barely above subsistence levels.
- Economic growth has been very uneven across time.
 - ▶ E.g. the productivity growth slowdown of 1990's
- Economic growth has been very uneven across countries.
 - ▶ E.g. growth miracles (Japan, South Korea, Singapore), growth disasters (Argentina, Sub-Saharan countries)

Economic Growth Over the Very Long Run

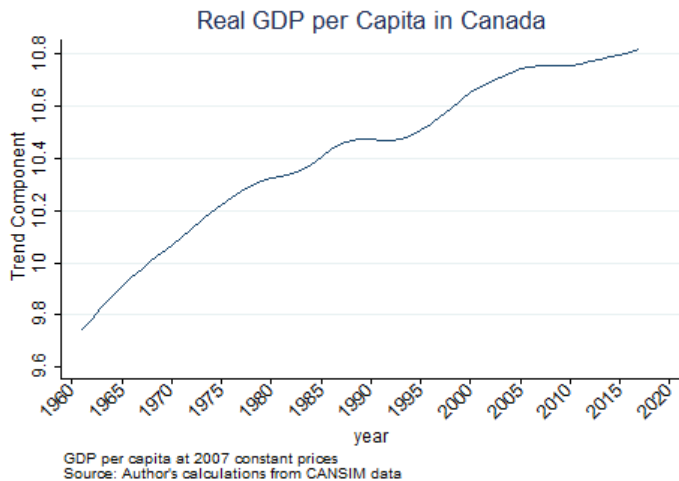
Economic Growth over the Very Long Run in Six Countries

Per capita GDP
(multiple of 300 dollars)



Source: Macroeconomics, Jones

Economic Growth Over Time



Defining Economic Growth

- Let g be a constant growth rate over a period of time, then ...

$$y_{t+T} = y_t(1 + g)^T$$

- Important properties of growth rates:

- (Rule of 70) Let $y_{t+T} = 2y_t$ and , then

$$T \approx \frac{0.7}{g}$$

- In general, the time it takes for a series to increase by a certain ratio does not depend on the initial level.
- Algebraic properties of growth rates (let the variable i grow at a constant growth rate g_i):

$$z = \frac{x}{y}$$

$$\Rightarrow g_z \approx g_x - g_y$$

$$z = x \cdot y$$

$$\Rightarrow g_z \approx g_x + g_y$$

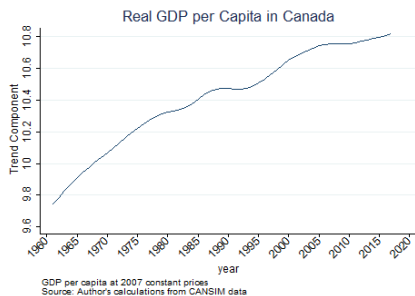
$$z = x^a \text{ (where } a \text{ is a constant)} \Rightarrow g_z \approx a \cdot g_x$$

GDP per capita trend revisited

- We can calculate average growth rates over a period of time T as follows:

$$\bar{g} = \left(\frac{y_{t+T}}{y_t} \right)^{\frac{1}{T}} - 1$$

- To separate the trend from the cycle, we choose a particular period of time and change the annual GDP per capita by that rate



The power of economic growth

Country	Period	Real GDP per Person at Beginning of Period*	Real GDP per Person at End of Period*	Growth Rate (per year)
United States	1870–2010	4 166	48 635	1.77
Indonesia	1900–2010	926	4 306	1.41
Pakistan	1900–2010	766	2 843	1.20
Bangladesh	1900–2010	648	1 854	0.96

Real GDP is measured in 2010 Canadian dollars.

Source: Mankiw, N., Kneebone, R., & McKenzie, K. (2017). *Principles of Macroeconomics* (7th Canadian ed.). Toronto, ON: Nelson Education

- At average growth rates it will take Bangladesh around 340 years, Pakistan around 237 years and Indonesia around 172 years to reach levels of real GDP per capita similar to US in 2010.

The power of economic growth

Country	Period	Real GDP per Person at Beginning of Period*	Real GDP per Person at End of Period*	Growth Rate (per year)
United States	1870–2010	4 166	48 635	1.77
Indonesia	1900–2010	926	4 306	1.41
Pakistan	1900–2010	766	2 843	1.20
Bangladesh	1900–2010	648	1 854	0.96

Real GDP is measured in 2010 Canadian dollars.

Source: Mankiw, N., Kneebone, R., & McKenzie, K. (2017). *Principles of Macroeconomics* (7th Canadian ed.). Toronto, ON: Nelson Education

- If average growth were to increase by a *quarter* of a percent, it would take Bangladesh around 270 years (diff. of 70 years), Pakistan around 196 years (diff. of 41 years) and Indonesia around 146 years (diff. of 26 years) to reach levels of real GDP per capita similar to US in 2010.